NOAH FOUDIL

Junior Industrial Engineer in Electronics and Embedded Systems

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PROFILE

Junior Industrial Engineer specialised in Electronics and Embedded Systems, with hands-on experience in embedded hardware, PCB design, FPGA/VHDL, sensor integration and system validation. Built and tested real-time embedded systems involving LiDAR, IMU and UAV platforms. Looking for a first engineering role in embedded electronics, test or system integration.

TECHNICAL SKILLS

Embedded Systems	STM32, ARM Cortex-M4, FreeRTOS, ESP32, AVR, Raspberry Pi, Embedded Linux, PWM, ADC, hardware/software interfacing
Digital Design & FPGA	VHDL, Intel Quartus Prime, Intel Cyclone V (DE0-CV), TimeQuest Timing Analyzer, finite-state machines, combinational and sequential logic
Electronics & PCB	KiCad, Altium, LTspice, Micro-Cap, analogue and digital circuit design, PCB layout, PCB assembly, debugging, signal integrity
Programming	C, C++, Python, MATLAB, VBA, Bash, Python `asyncio`, multithreaded programming
Sensors & Signal Processing	LiDAR, ToF, IMU, sensor fusion, Extended Kalman Filter (EKF), DFT, attitude estimation, point-cloud processing
Communication Protocols	UART, SPI, I^2C, CAN, UDP, Wi-Fi, BLE, WebSockets, MAVLink, CRC-8 validation
Validation & Test	Requirements-driven testing and validation, test benches, oscilloscope diagnostics, hardware validation, static timing analysis

PROJECTS

Master's Thesis: Indoor UAV Environmental Mapping using 2D LiDAR *Dec 2025 – Mar 2026*
Research Internship, BioVLSI Lab, National Yunlin University of Science and Technology (YunTech), Taiwan

- Set measurable targets for cost, mass, flight endurance, LiDAR communication integrity and reconstruction accuracy; met 4 of 5 targets, including a material cost of EUR 269, an all-up mass of 910 g and 7 minutes of flight endurance.
- Integrated a Quad-X UAV platform built around a Matek F405 flight controller running ArduPilot EKF3, a Raspberry Pi Zero 2 W companion computer, LiDAR and ToF sensors.
- Implemented a four-task concurrent architecture in Python using `asyncio` on Embedded Linux for MAVLink telemetry, ToF ranging, LiDAR UART parsing at 230,400 bps, CRC-8 validation and WebSocket broadcasting.
- Investigated a LiDAR throughput bottleneck of 2,245 points/s versus 4,500 points/s nominal; traced it to cooperative scheduling desynchronisation and proposed a dedicated I/O thread architecture.
- Built a pseudo-3D point-cloud reconstruction method using attitude interpolation, deskewing and voxel-grid filtering; measured sensor-level accuracy of 10–35 mm against a 55 mm target.

HVDC Capacitor Test Bench *Sep 2023 – Oct 2023*
Engineering Internship, Hitachi Energy, Charleroi, Belgium

- Contributed to the build, wiring and commissioning of an automated HVDC capacitor test bench, with a focus on operator safety, measurement integrity and system reliability.
- Simulated and sized cascaded passive RC filtering stages in LTspice to reduce DC power-supply ripple and improve leakage-current measurement fidelity.
- Automated the extraction, filtering and centralisation of Fluke measurement data using VBA in Excel, reducing repetitive manual processing and standardising fault-detection workflows.

Precision LCR Meter Design *2025*
Master's Coursework Project, Independent Design Project

- Created an auto-balancing-bridge LCR meter targeting measurement error below 1% for impedance, capacitance, inductance, quality factor and dissipation factor characterisation.
- Defined an ESP32-based acquisition architecture using two external 16-bit ADCs for simultaneous voltage and current sampling, enabling DFT-based impedance extraction without time-multiplexing phase errors.
- Implemented analogue signal-conditioning stages using INA128, PGA103, DAC8811 and OPA192; routed a KiCad PCB with a continuous ground plane and analogue/digital separation.
- Assembled and tested the hardware stage by stage using oscilloscope diagnostics and systematic fault isolation.

FPGA Game Engine on Intel Cyclone V, VHDL *Oct 2025*
HELHa Coursework, Team of 3

- Co-created a real-time Flappy Bird-style game in VHDL for an Intel Cyclone V FPGA (DE0-CV), displayed across six seven-segment displays.
- Created eight independent VHDL modules and dedicated test benches; achieved a timing-clean Quartus implementation at approximately 155 MHz for a 50 MHz system clock while using approximately 1% of logic resources.

EDUCATION

Master's in Industrial Engineering, Electronics Specialisation *2024 – 2026*
Haute École Louvain en Hainaut (HELHa), Mons, Belgium *Expected 2026*

Bachelor's in Industrial Engineering *2020 – 2024*
Haute École Louvain en Hainaut (HELHa), Mons, Belgium

ADDITIONAL INFORMATION

Languages	French: Native
	English: C1 reading and listening; able to work with technical documentation and discussions
	Dutch: Basic knowledge, secondary-school immersion programme
	Mandarin Chinese: Basic knowledge
Driving licence	Category B | Personal vehicle: Available
Availability	Available immediately